

1 Title

We have the error function $e(z')$, where z' is the pint in image coordinates. We can describe the error as

$$e^i(x) = d(\Pi(kx_p^i)) \quad (1.1)$$

Let's consider the iso-level lines of point A and point B, if we image a point through equation 1, where point A is the center and it's an obstacle. If the distance of the evaluated point B is 0, then the distance from A and B is 0: the point B is on the obstacle.

The derivative of the error function is

$$d(\Pi(K((xboxplus\Delta x)p))) \quad (1.2)$$

let's expand the definition of boxplus:

$$d\{\Pi[k(v2t(\Delta x)x_p)]\} \quad (1.3)$$

. We now have the derivative of the distance map in the evaluated pixel (this is a 2D since the distance map is a $R^2 \rightarrow R^1$ function)

$$\frac{\partial d}{\partial p_{img}}|_{p_{im}=\Pi(kx_p^i)} * \frac{\partial \Pi}{\partial u,v}|_{u,v=(kx_p^i)} * K * J_{icp} \quad (1.4)$$

where $p_{img} = \Pi(kx_p^i)$ is a 2D vector. Dimensions:

- $\frac{\partial d}{\partial p_{img}}|_{p_{im}=\Pi(kx_p^i)}$ is a 1×2 ,
- $\frac{\partial \Pi}{\partial u,v}|_{u,v=(kx_p^i)}$ is 2×3 ,
- k is 3×3 ,
- J_{ICP} is 3×6 .

we obtain a vector

2 Notation

d represent the distance map.

Π is the *proj* (projection).